

Benjamin Farsijani

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EDUCATION

University of Ottawa — Honours B.Sc., Computer Science Sep 2023 - May 2027

GPA: 3.7 / 4.0 • **Coursework:** Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Systems Programming

TECHNICAL SKILLS

Languages: Python, Go, C/C++, Rust, TypeScript/JS, SQL, Bash

ML & Data: PyTorch, XGBoost, scikit-learn, NumPy, pandas, NetworkX; time-series modeling, walk-forward CV, anomaly detection, graph ML (Louvain, PageRank, HITS), Model Predictive Control, stochastic simulation, Monte Carlo methods

Systems & Infra: Linux/POSIX, multithreading, low-latency networking, TCP/IP, epoll, Docker, PostgreSQL, SQLite, Flask, REST/WebSocket

Embedded: ESP32 firmware, I2C/SPI/UART, sensor integration, hardware bring-up, 3D printing

PROJECTS

Benny — Algorithmic Crypto Trading System

Production

Python, XGBoost, CCXT, Binance API, PostgreSQL, Docker, Flask

- Built end-to-end ML trading system ingesting real-time market data from Binance across 6 liquid assets (BTC, ETH, SOL, BNB, XRP, DOGE); runs continuously in production.
- Engineered 50+ features with strict zero-lookahead construction: GARCH(1,1) volatility, Parkinson high-low range, VWAP deviation, OBV, ADX, Stochastic RSI, multi-timeframe log returns.
- Trained XGBoost directional classifier validated via 5-fold walk-forward CV; reported per-fold Sharpe, max drawdown, hit rate, and simulated P&L vs buy-and-hold baseline.
- Designed full risk layer (probability-gated entries, 3% TP / 1.5% SL, 0.1% fee modeling, per-symbol sizing across \$6K paper portfolio); deployed live containerized dashboard tracking equity curve and trade history.

Azadi — Threat Network Intelligence Dashboard

In Development

Python, NetworkX, GDEL T, Dash, Cytoscape, SQLite

- Building Palantir Gotham-style intelligence platform analyzing actor dynamics during the Iranian Revolution (1979-1981) from GDEL T event data.
- Ingested GDEL T 1.0 bulk data into a directed weighted NetworkX graph; nodes typed by CAMEO actor codes across Government, Military, Religious, and Opposition factions.
- Implemented Louvain community detection (20 communities, modularity 0.10) and composite influence scoring (PageRank + HITS); daily z-score anomaly detection surfaced 15 event spikes including the Nov. 4, 1979 US Embassy hostage crisis.

Celsius — IoT Environmental Monitoring System

1st Place, Design Day

ESP32, C/C++, React Native, BME680, MAX9814, ML

- Designed wall-mounted IoT system for Garderie Bernadette Child Care Centre at the University of Ottawa; ran continuously through pilot deployment, currently paused pending the centre's relocation.
- Integrated BME680 (temp/humidity/VOC) and MAX9814 (sound) over I2C with WiFi streaming, 3D-printed lithium-battery-powered enclosure, and React Native app; ML anomaly-detection layer flagging air-quality dips, noise spikes, and temperature excursions with real-time alerts.

Stochastic Control & MPC for Thermal Oil Recovery

Research

Python, PyTorch, CasADi, NumPy

- Developed stochastic simulation environment modeling delayed nonlinear dynamics (steam injection → reservoir temperature → production output); generated 100k+ synthetic multivariate observations.
- Implemented constrained MPC with a learned PyTorch state-space model solved in CasADi, targeting a projected 5-10% reduction in steam-to-oil ratio vs heuristic baselines.

EXPERIENCE

Technical Mentor, Embedded Systems & Prototyping — CEED, University of Ottawa

Sep 2024 - May 2025

- Mentored engineering students through firmware debugging, peripheral integration (I2C, SPI, UART), and hardware bring-up in CEED's Makerspace; guided hardware/software co-design across multiple student teams under tight competition deadlines.

Founder — Benny Blends Barbershop

2018 - Present

- Built and ran a barbershop from a home studio while balancing school full-time; \$25,000+ cumulative revenue (\$15K peak year), thousands of clients served, all operations owned independently.

Competitive Soccer Coach — Ottawa South United

2020 - 2025

- Coached 500+ youth athletes across competitive recreational programs over five seasons.